# Coding & Solution

## SMARTBRIDGE - Coding & Solution

- **Date:** 21/09/2026
- **Team ID:** SWTID-2026-7817
- **Team Size:** 4
- **Team Leader:** Karan S
- **Team Members:** SANJAY .M, Santhosh S, Yaseen P
- **Project Name:** AI FitTrack
- **Maximum Marks:** 5 Marks

## Solution Summary

| Field | Details |
|---|---|
| **Repository Link / URL** | [https://drive.google.com/drive/folders/1hwmKxkHQkUI62JR1svI3xCQ2ShZ6Lj9y](https://drive.google.com/drive/folders/1hwmKxkHQkUI62JR1svI3xCQ2ShZ6Lj9y) |
| **Programming Language(s)** | JavaScript (Node.js runtime environment v16+) |
| **Framework(s) Used** | Express.js (REST API framework), Mongoose ODM |
| **Key Features Implemented** | • JWT User Registration, Login & Profile Retrieval.<br><br>• Full Workout CRUD Management (Create, Read, Search, Update, Delete). |

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|---|---|
| | • Google Gemini AI Workout Recommendations & Performance Insights.<br><br>• Centralized JSON Error Handling & Schema Validations. |
| **Pending / Incomplete Features** | None. All planned backend REST endpoints and AI features are 100% operational. |
| **Setup / Run Instructions** | 1. Clone repository and navigate into project directory: cd Code Files.<br><br>2. Install dependencies: npm install.<br><br>3. Configure .env file with PORT, MONGO_URI, JWT_SECRET, and GEMINI_API_KEY.<br><br>4. Start server: npm start.<br><br>5. Execute API tests via Thunder Client / Postman at http://localhost:5000. |

## Code Quality Checklist

| S.No | Criteria PDF | Status (Yes / No) PDF |
|---|---|---|

| | | |
|---|---|---|
| **1** | **Code is modular and organized into functions / classes** | **Yes** |
| **2** | **Meaningful variable and function names are used** | **Yes** |
| **3** | **Code includes comments / documentation where necessary** | **Yes** |
| **4** | **Error handling is implemented for critical operations** | **Yes** |
| **5** | **The application runs without critical errors** | **Yes** |
| **6** | **Code is committed to a version control repository** | **Yes** |

**Additional Notes / Comments:** Application verified successfully using Thunder Client. API endpoints handle database operations and Gemini AI requests within target response times.